Insurance Policy Sales and Claims Report

Objective

The objective of this assignment was to create a large-scale insurance dataset and analyse profitability, risk patterns, and claim trends using structured analytical queries.

Created a synthetic dataset representing 1000000 car insurance policies sold during 2024. Claims activity was simulated for 2025 and early 2026.

Data Simulation Approach

Two datasets were created:

Policy sales dataset: The policy sales dataset contains 1000000 records, each representing a policy purchased in 2024.

Key Assumptions:

- Vehicle Value: 100000
- Premium: 100 p.a. of policy tenure
- Policies start 365 days after the purchase
- Policies sold evenly throughout 2024

Policy Tenure Distribution:

Tenure	Distribution
1 year	20%
2 year	30%
3 year	40%
4 year	10%

Columns:

- Customer_ID
- Vehicle_ID
- Vehicle_Value
- Premium
- Policy_Purchase_Date
- Policy_Start_Date
- Policy_End_Date
- Policy_Tenure

Claims dataset: It was created by the following assumptions

Claims in 2025

- Vehicles purchased on 7th, 14th, 21st, 28th of each month.
- 30% of those vehicles filed claims.
- Claims occurred exactly on the policy start date.
- Each vehicle could file one claim per year.

Claims in 2026

- Period: Jan 1- Feb 28 2026.
- 10% of vehicles with 4-year tenure filed claims.
- Claims distributed evenly across the days.
- Vehicles that claimed in 2025 could claim again in 2026.

Claim Amount assumed as: 10% of vehicle value = 10000.

Analytical Queries and Results

1. Total premium collected in 2024

Premium= 100 * Policy Tenure

```
SELECT SUM(Premium) as Total_Premium
FROM policy_sales;
```

Output:

	Total_Premium
▶	240000000

Total Premium collected = 240,000,000

2. Total claim cost per year

Claims occurred in 2025.

The total claim cost was calculated by aggregating all claim amounts by year and month.

```
SELECT YEAR(Claim_Date) as years,
MONTH(Claim_Date) as months,
SUM(Claim_Amount) as Total_Claim_Cost
FROM claims_data
GROUP BY years, months
ORDER BY years, months;
```

Output:

	years	months	Total_Claim_Cost
▶	2025	1	33060000
	2025	2	32430000
	2025	3	33140000
	2025	4	32860000
	2025	5	32520000
	2025	6	32750000
	2025	7	32300000
	2025	8	32390000
	2025	9	33280000
	2025	10	33610000
	2025	11	33120000
	2025	12	32450000

3. Claim cost to premium ratio for each policy tenure (1, 2, 3, and 4 years).

```
SELECT p.Policy_Tenure,
SUM(c.Claim_Amount)/ SUM(Premium) AS claim_ratio
FROM policy_sales p
JOIN claims_data c
ON p.Vehicle_ID = c.Vehicle_ID
GROUP BY p.Policy_Tenure
ORDER BY Policy_Tenure;
```

Output:

	Policy_Tenure	claim_ratio
▶	1	100
	2	50
	3	33.333333333333336
	4	25

Longer tenure policies generally improve profitability because **premium revenue accumulates over longer durations while claims remain limited**.

4. Claim ratio by policy purchase month

Claim ration was analysed by the month in which policies were purchased.

Since policy start dates occur exactly one year after purchase, claim patterns follow similar monthly patterns.

```sql
SELECT MONTH(p.Policy_Purchase_Date) as sale_month,
SUM(c.Claim_Amount)/SUM(p.Premium) as Ratio
FROM policy_sales p
JOIN claims_data c
ON p.Vehicle_ID= c.Vehicle_ID
GROUP BY sale_month
ORDER BY sale_month;
```

Output:

	sale_month	Ratio
▶	1	41.63203626747261
	2	41.82896943118793
	3	41.8381517485166
	4	41.57914715930659
	5	41.7887432536623
	6	41.51876267748479
	7	41.15698267074414
	8	41.782765737874094
	9	41.5792103948026
	10	41.98101423932051
	11	41.70234197935029
	12	41.9304819744153

5. Potential future claim liability

Assuming every vehicle that has not yet made a claim eventually files exactly one claim, the future claim liability was estimated.

Claim amount per vehicle: 10000

```sql
SELECT (COUNT(*) -COUNT(DISTINCT c.Vehicle_ID)) * 10000 AS potential_claim_liability
FROM policy_sales p
LEFT JOIN claims_data c
ON p.Vehicle_ID= c.Vehicle_ID
;
```

Output:

	potential_claim_liability
▶	9606090000

The maximum potential future claim is 9,606,090,000, if all the remaining vehicles file claims during their policy tenure.

6. I) Premium earned until Feb 28, 2026

```
SELECT SUM(
    (Premium/(Policy_Tenure * 365)) * DATEDIFF(LEAST('2026-02-28', Policy_End_Date), Policy_Start_Date)
) as premium_earned
FROM policy_sales
WHERE Policy_Start_Date <= '2026-02-28';
```

Output:

	premium_earned
▶	65889852.1509

The premium earned up to **Feb 28, 2026: 65,889,852.15.**

II) Future premium revenue

```
SELECT (SUM(Premium) - SUM(
    (Premium/(Policy_Tenure * 365)) * DATEDIFF(LEAST('2026-02-28', Policy_End_Date), Policy_Start_Date)
))/46 as expected_monthly_premium
FROM policy_sales
WHERE Policy_Start_Date <= '2026-02-28';
```

Output:

	expected_monthly_premium
▶	3785003.21411164

Future revenue: 3,785,003.21

Bonus Analysis

1. Most profitable policy tenure

Profitability was evaluated using loss ratio:

Loss ratio = total claims / total premium

```
SELECT p.Policy_Tenure,
SUM(p.Premium) as total_premium,
SUM(c.Claim_Amount) as total_claims,
SUM(c.Claim_Amount)/ SUM(p.Premium) as loss_ratio
FROM policy_sales p
LEFT JOIN claims_data c
ON p.Vehicle_ID = c.Vehicle_ID
GROUP BY p.Policy_Tenure
ORDER BY loss_ratio;
```

Output:

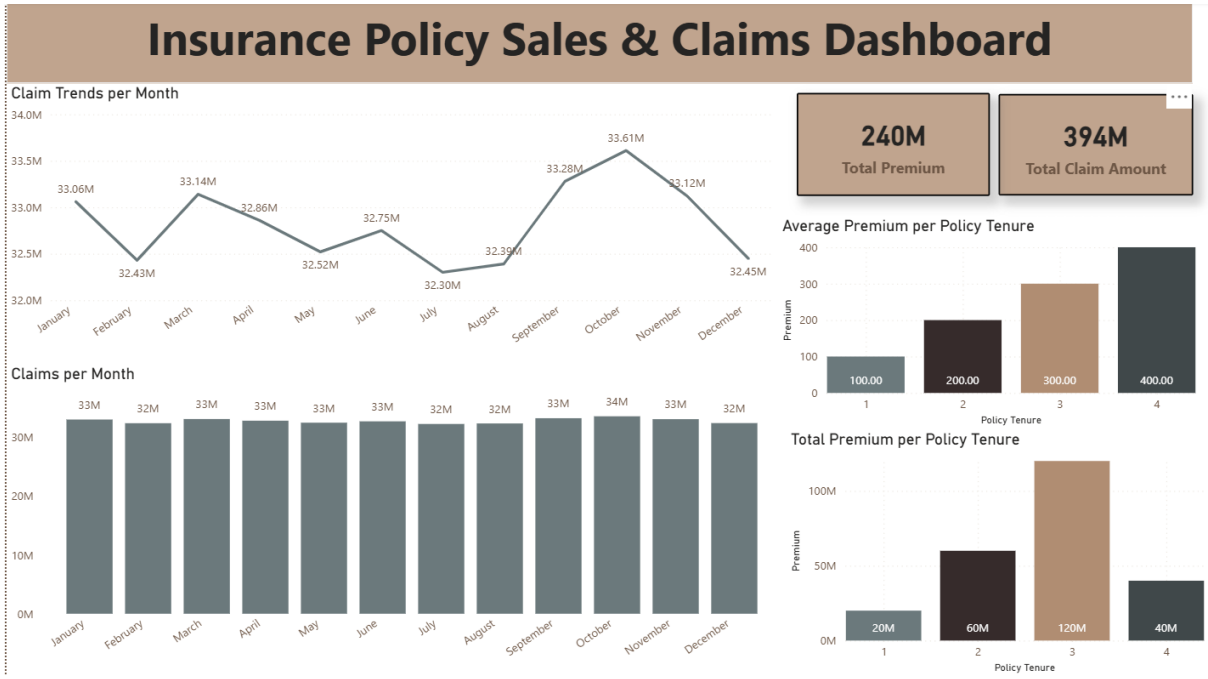
Policy_Tenure	total_premium	total_claims	loss_ratio
4	40000000	39590000	0.98975
3	120000000	156720000	1.306
2	60000000	118670000	1.9778333333333333
1	20000000	78930000	3.9465

Policies with longer tenure appear more profitable because:

- Higher total premium
- Claims do not increase proportionally
- Risk is spread across a longer period of time

Therefore, encouraging longer-tenure policies may improve profitability.

2. Monthly claim trends



3. Portfolio Loss ratio

```
SELECT SUM(c.Claim_Amount)/SUM(p.Premium) AS portfolio_loss_ratio
FROM policy_sales p
LEFT JOIN claims_data c
ON p.Vehicle_ID=c.Vehicle_ID
;
```

Output:

	portfolio_loss_ratio
▶	1.6412916666666666

Loss Ratio: 1.6413 (164.13%)

A loss ratio of 164% means for every **100 in premium collected**, the insurer pays **164 in claims.**

This indicates that the portfolio is **unprofitable under the current pricing assumptions**.

4. Impact of 5% increase in claim frequency

If claim frequency increases by 5% annually, total claim costs would increase proportionally.

```
SELECT COUNT(*)/ ( SELECT COUNT(*) FROM policy_sales) AS current_claim_frequency
FROM claims_data
;

SELECT SUM(Claim_Amount) AS current_claim_cost,
SUM(Claim_Amount) * 1.05 as projected_claim_cost
FROM claims_data;

SELECT SUM(c.Claim_Amount)/ SUM(p.Premium) as current_loss_ratio,
(SUM(c.Claim_Amount)*1.05)/ SUM(p.Premium) as projected_loss_ratio
FROM policy_sales p
LEFT JOIN claims_data c
ON p.Vehicle_ID=c.Vehicle_ID
;
```

Output:

	current_claim_frequency
▶	0.0394

	current_claim_cost	projected_claim_cost
▶	393910000	413605500

	current_loss_ratio	projected_loss_ratio
▶	1.6412916666666666	1.72335625

Impact:

- Higher claim costs
- Increase loss ratio
- Reduced insurer profitability

Without adjustments to premium pricing or risk underwriting, rising claim frequency could significantly worsen financial performance.

Business Insights

The analysis reveals several important findings:

- The current premium structure is **significantly underpriced relative to the claim risk**.
- Claim events are concentrated around specific policy start dates due to simulation assumptions.
- Longer-tenure policies provide **more stable revenue**.
- Future claim liability represents a significant financial exposure if all vehicles eventually file claims.
- Rising the claim frequency could substantially reduce profitability.

Recommendations

Based on the analysis, the insurer should consider:

- Increasing premium pricing to better reflect claim risk.
- Encouraging longer policy tenures.
- Maintaining adequate claim reserves.